Introduction to Computing

Lecture 8

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Practice Questions

Questions

Write a C++ program that prompts the user to input two integer values and find the greatest value of the two values.

Solution

```
#include <iostream>
using namespace std;

int main()
{
    int a, b;
    cin >> a;
    cin >> b;

    if (a > b)
    {
        cout << "Max: " << a; //output max
    }

    else
    {
        cout << "Max: " << b; //output max
    }
    return 0;
}
```

Questions

Write a C++ program that prompts the user to input three integer values and find the greatest value of the three values.

Solution

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;
    cin >> a;
    cin >> b;
    cin >> c;

    if ((a > b) && (a > c))
        cout << "Max: " << a; //output max

    else if ((b > a) && (b > c))
        cout << "Max: " << b; //output max

    else
        cout << "Max: " << c; //output max

    return 0;
}
```

Solution

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c, max;
    cin >> a;
    cin >> b;
    cin >> c;

    max = a; //let max take the first value

    if (max < b)
        max = b; // compare max with b and update max

    if (max < c)
        max = c; //compare max with c and take c

    cout << "Max: " << max; //output max
    return 0;
}
```

Questions

Write a C++ program that tells whether the input number is even or odd.

Solution

```
#include<iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter a number :";
    cin >> num;

    if (num % 2 == 0)
    {
        cout << "This is an even number";
    }
    else
    {
        cout << "This is an odd number";
    }

}
```

Questions

Write a C++ program that tells whether the input number is even or odd using `Switch`.

Solution

```
#include<iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter a number :";
    cin >> num;
    int choice = num % 2;

    switch (choice)
    {
    case 0:
        cout << "This is an even number";
        break;
    case 1:
        cout << "This is an odd number";
        break;
    }

}
```

Questions

Write a method with a while loop to prints n through 0 in square brackets. For example, if n = 6 print

Please enter number: <6>
[6] [5] [4] [3] [2] [1] [0]

Solution

```
#include <iostream>
using namespace std;

int main()
{
    int n;

    cout << "Please enter number :";
    cin >> n;

    int i = n;

    while (i >= 0)
    {
        cout << "[" << i << "] ";
        i--;
    }

    return 0;
}
```

Questions

Write a C++ program that prompts the user to input like this:

1/2

and your program output:

0.5

Solution

```
#include<iostream>
using namespace std;
int main()
{
    float nominator;
    float denominator;
    char useless;

    cout << "Enter expression :";
    cin >> nominator >> useless >> denominator;
    cout << "Answer :"<< nominator/denominator;
}
```

ASCII Values

ASCII control characters		
00	NULL	(Null character)
01	SOH	(Start of Header)
02	STX	(Start of Text)
03	ETX	(End of Text)
04	EOT	(End of Trans.)
05	ENQ	(Enquiry)
06	ACK	(Acknowledgement)
07	BEL	(Bell)
08	BS	(Backspace)
09	HT	(Horizontal Tab)
10	LF	(Line feed)
11	VT	(Vertical Tab)
12	FF	(Form feed)
13	CR	(Carriage return)
14	SO	(Shift Out)
15	SI	(Shift In)
16	DLE	(Data link escape)
17	DC1	(Device control 1)
18	DC2	(Device control 2)
19	DC3	(Device control 3)
20	DC4	(Device control 4)
21	NAK	(Negative acknowl.)
22	SYN	(Synchronous idle)
23	ETB	(End of trans. block)
24	CAN	(Cancel)
25	EM	(End of medium)
26	SUB	(Substitute)
27	ESC	(Escape)
28	FS	(File separator)
29	GS	(Group separator)
30	RS	(Record separator)
31	US	(Unit separator)
127	DEL	(Delete)

ASCII printable characters					
32	space	64	@	96	`
33	!	65	A	97	a
34	"	66	B	98	b
35	#	67	C	99	c
36	$	68	D	100	d
37	%	69	E	101	e
38	&	70	F	102	f
39	'	71	G	103	g
40	(	72	H	104	h
41	)	73	I	105	i
42	*	74	J	106	j
43	+	75	K	107	k
44	,	76	L	108	l
45	-	77	M	109	m
46	.	78	N	110	n
47	/	79	O	111	o
48	0	80	P	112	p
49	1	81	Q	113	q
50	2	82	R	114	r
51	3	83	S	115	s
52	4	84	T	116	t
53	5	85	U	117	u
54	6	86	V	118	v
55	7	87	W	119	w
56	8	88	X	120	x
57	9	89	Y	121	y
58	:	90	Z	122	z
59	;	91	[	123	{
60	<	92	\	124	|
61	=	93	]	125	}
62	>	94	^	126	~
63	?	95	_		

Extended ASCII characters							
128	Ç	160	á	192	└	224	Ó
129	ü	161	í	193	┴	225	ß
130	é	162	ó	194	┬	226	Ô
131	â	163	ú	195	├	227	Ò
132	ä	164	ñ	196	─	228	õ
133	à	165	Ñ	197	┼	229	Õ
134	å	166	ª	198	ã	230	µ
135	ç	167	º	199	Ã	231	þ
136	ê	168	¿	200	╚	232	Þ
137	ë	169	®	201	╔	233	Ú
138	è	170	¬	202	╩	234	Û
139	ï	171	½	203	╦	235	Ù
140	î	172	¼	204	╠	236	ý
141	ì	173	¡	205	═	237	Ý
142	Ä	174	«	206	╬	238	¯
143	Å	175	»	207	¤	239	´
144	É	176	░	208	ð	240	≡
145	æ	177	▒	209	Ð	241	±
146	Æ	178	▓	210	Ê	242	‗
147	ô	179	│	211	Ë	243	¾
148	ö	180	┤	212	È	244	¶
149	ò	181	Á	213	ı	245	§
150	û	182	Â	214	Í	246	÷
151	ù	183	À	215	Î	247	¸
152	ÿ	184	©	216	Ï	248	°
153	Ö	185	╣	217	┘	249	¨
154	Ü	186	║	218	┌	250	·
155	ø	187	╗	219	█	251	¹
156	£	188	╝	220	▄	252	³
157	Ø	189	¢	221	¦	253	²
158	×	190	¥	222	Ì	254	■
159	ƒ	191	┐	223	▀	255	nbsp

Close

Questions

Write program to check whether a character is alphabet or not

Solution

```
#include <iostream>
using namespace std;

int main()
{
    char ch;
    cin >> ch;

    if ((ch >= 97 && ch <= 122) || (ch >= 65 && ch <= 90))
    {
        printf("Character is an ALPHABET.");
    }
    else
    {
        printf("Character is NOT ALPHABET.");
    }

    return 0;
}
```

Questions

Write program to which takes alphabet as an input and print next alphabet

Solution

```
#include <iostream>
using namespace std;

int main()
{
    char ch;

    cin >> ch;

    cout << "Next character is: " << ++ch;

    return 0;
}
```

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over